H3 SF2525

Tommaso Brambilla

brambi@kth.se

Exercise 1

1.a

Given a stock that solves the Ito stochastic differential equation:

$$dS(t) = rS(t)dt + \sigma S(t)dW(t)$$

$$S(0) = S_0$$

we already know that it is solved by:

$$S(T) = exp((r - \sigma^2/2)T + \sigma W(T))S_0$$

Now we want to simulate the price of a European call option, defined as:

$$f(0, S_0) = e^{-rT}\mathbb{E}[max(S(T) - K, 0)|S(0) = S_0]$$

by a Monte Carlo method.
This means that we have to approximate the expected value in the following way:

$$\mathbb{E}[max(S(T) - K, 0)|S(0) = S_0] \approx \frac{1}{N}\sum_{i=1}^{N} max(S(T, w_i) - K, 0)$$

Thus, the approximation's error consists of:

$$\epsilon_N = e^{-rT}(\sum_{i=1}^{N} \frac{\mathbb{E}[max(S(T) - K, 0)|S(0) = S_0] - max(S(T, w_i) - K, 0)}{N})$$

Since the expected value of the numerator is 0 and each sample $S(T, w_i)$ is independent, identically distributed, we can apply the Central Limit Theorem and state that:

$$\sqrt{N}\epsilon_N \to \sigma\nu \qquad as \;\; N \to \infty$$

where N is the number of samples, σ is the square root of the variance of the error and ν is a random variable normally distributed with expected value 0 and variance 1 (note that we can approximate ν as the square root of the sample variance $\hat{\sigma^2}$

$$\hat{\sigma}^2 = \frac{1}{N}\sum_{i=1}^{N}((max(S(T, w_i) - K, 0) - \sum_{m=1}^{N} \frac{max(S(T, w_m) - K, 0)}{N})^2$$

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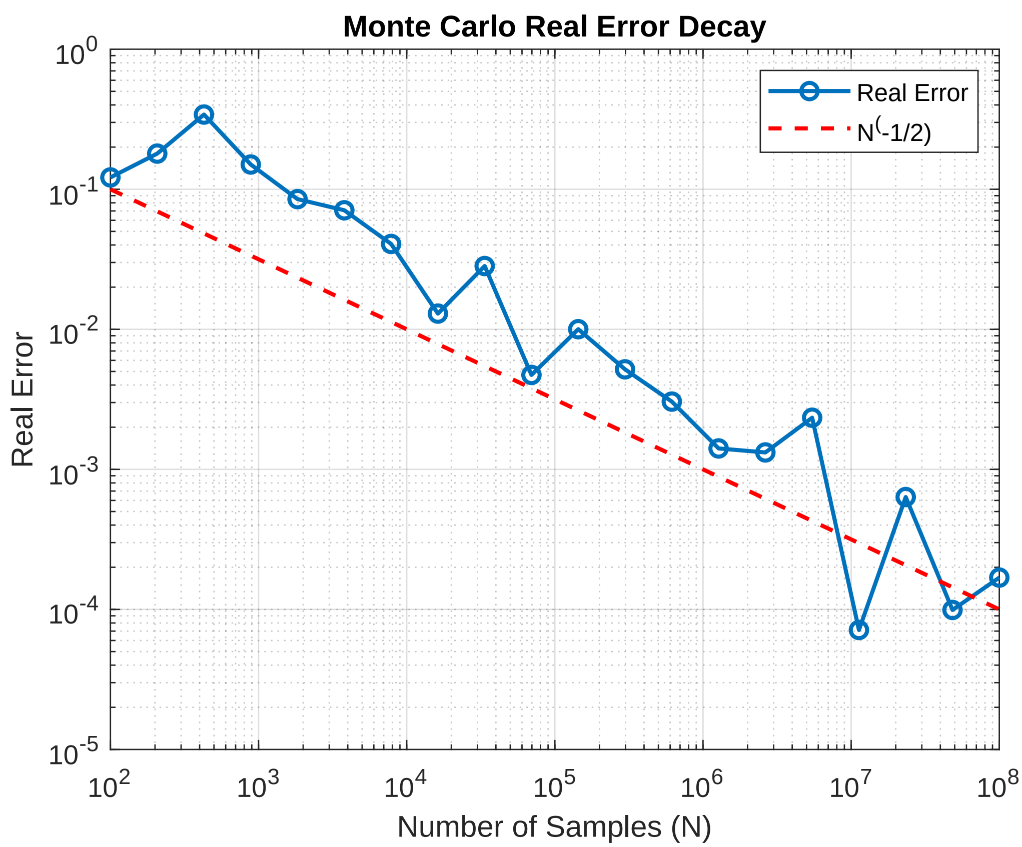


Monte Carlo Real Error Decay
Real Error
N^(-1/2)
10^0
10^-1
10^-2
10^-3
10^-4
10^-5
10^2
10^3
10^4
10^5
10^6
10^7
10^8
Real Error
Number of Samples (N)

1.b

Now we want to compute the corresponding sensitivity with the finite difference quotient:

$$\Delta \approx \frac{f(0, s + \Delta s) - f(0, s)}{\Delta s}$$

We will use the same Monte Carlo approximation as in point 1.a:

$$f(0, S_0) = e^{-rT}\mathbb{E}[max(S(T) - K)|S(0) = S_0] \approx \sum_{j=1}^{N} \frac{e^{-rT} max(S(T, w_j) - K, 0)}{N}$$

$$f(0, S_0 + \Delta S) = e^{-rT}\mathbb{E}[max(S(T) - K)|S(0) = S_0 + \Delta S] \approx \sum_{j=1}^{N} \frac{e^{-rT} max(S_{\Delta S}(T, w_j) - K, 0)}{N}$$

where

$$S(T, w_j) = exp((r - \sigma^2/2)T + \sigma W(T))S_0$$
$$S_{\Delta S}(T, w_j) = exp((r - \sigma^2/2)T + \sigma W(T))(S_0 + \Delta S)$$

Here we can apply the Central Limit Theorem to compute the error $\epsilon = \frac{\sum_{j=1}^{N} e^{-rT} max(S_{\Delta S}(T,w_j)-K,0) - e^{-rT} max(S(T,w_j)-K,0)}{N\Delta s}$ and after some computations we get:

$$\sqrt{N}\epsilon \to \sigma\nu$$

as $N \to \infty$ where ν is normally distributed with zero mean and variance one.
If we want to find a better method to solve this problem, we can choose another finite difference method to compute the approximation of the sensitivity:

$$\Delta \approx \frac{f(0, s + \Delta s) - f(0, s - \Delta s)}{2\Delta s}$$

Exercise 2

2.a

Now we consider the following stochastic differential equation

$$dY(t) = (-\alpha(2 + Y(t)) + 0.4\sqrt{\alpha}\sqrt{1 - \rho^2})dt + 0.4\sqrt{\alpha}d\hat{Z(t)}$$

where W and Z are independent Wiener processes, $\alpha > 0$ and

$$\hat{Z}(t) = \rho W(t) + \sqrt{1 - \rho^2}Z(t)$$

In order to solve it in a closed form, we first notice that the term $Y(t)$ appears in the deterministic part only (there is not $Y(t)$ in the coefficient of $d\hat{Z}(t)$), so we can solve it using the integrator factor method.
In particular we rearrange the equation leaving on the right hand side all the terms independent of $Y(t)$, then we multiply each term by the integrating factor:

$$\exp\left(\int_0^t \alpha\, ds\right) = e^{\alpha t}$$

in order to write the LHS as the differential of a product. Then we obtain the solution $Y(t)$ integrating each term over the interval $[0, t]$ and dividing by $e^{\alpha t}$.
Here are the computations:

$$dY(t) = (-\alpha(2 + Y(t)) + 0.4\sqrt{\alpha}\sqrt{1 - \rho^2})dt + 0.4\sqrt{\alpha}d\hat{Z(t)}$$

$$\implies dY(t) + \alpha Y(t)dt = (-\alpha 2 + 0.4\sqrt{\alpha}\sqrt{1-\rho^2})dt + 0.4\sqrt{\alpha}d\hat{Z(t)}$$
$$\implies e^{\alpha t}dY(t) + e^{\alpha t}\alpha Y(t)dt = e^{\alpha t}(-\alpha 2 + 0.4\sqrt{\alpha}\sqrt{1-\rho^2})dt + e^{\alpha t}0.4\sqrt{\alpha}d\hat{Z(t)}$$
$$\implies d(e^{\alpha t}Y(t)) = e^{\alpha t}(-\alpha 2 + 0.4\sqrt{\alpha}\sqrt{1-\rho^2})dt + e^{\alpha t}0.4\sqrt{\alpha}d\hat{Z(t)}$$
$$\implies e^{\alpha t}Y(t) = Y_0 + \int_0^t e^{\alpha s}(-\alpha 2 + 0.4\sqrt{\alpha}\sqrt{1-\rho^2})\, ds + \int_0^t e^{\alpha s}0.4\sqrt{\alpha}d\hat{Z(s)}\, ds$$

Finally obtaining:

$$Y(t) = e^{-\alpha t}Y_0 + (-2 + \frac{0.4\sqrt{1-\rho^2}}{\sqrt{\alpha}})(1 - e^{-\alpha t}) + \int_0^t e^{-\alpha(t-s)}0.4\sqrt{\alpha}d\hat{Z(s)}\, ds$$

Now we want to compute $\mathbb{E}[Y(t)]$, $\text{Var}(Y(t))$ and their limits as $t \to \infty$
To compute $\mathbb{E}[Y(t)]$, we take the expected value of LHS and RHS and consider that

$$\mathbb{E}[\int_0^t e^{-\alpha(t-s)}0.4\sqrt{\alpha}d\hat{Z(s)}\, ds] = 0$$

since $\mathbb{E}[d\hat{Z}(s)] = 0$ (because $d\hat{Z}(s) = \rho dW(s) + \sqrt{1-\rho^2}dZ(s)$ and $\mathbb{E}[dW(s)] = \mathbb{E}[dZ(s)] = 0$ from the definition of Wiener processes. Thus:

$$\mathbb{E}[Y(t)] = e^{-\alpha t}Y_0 + (-2 + \frac{0.4\sqrt{1-\rho^2}}{\sqrt{\alpha}})(1 - e^{-\alpha t})$$

taking the limit as $t \to \infty$:

$$\mathbb{E}[Y(t)] \to -2 + \frac{0.4\sqrt{1-\rho^2}}{\sqrt{\alpha}} \approx -2 + \frac{0.3816}{\sqrt{\alpha}}$$

Then we can compute the variance using this property:

$$\text{Var}(Y(t)) = \mathbb{E}[Y^2(t)] - \mathbb{E}[Y(t)]^2$$

After some computations we got:

$$\text{Var}(Y(t)) = 0.08(1 - e^{-2\alpha t}) \to 0.08$$

The interpretation of these results is as follows:
the expected value of $Y(t)$ tends to stabilize around the value $-2 + \frac{0.3816}{\sqrt{\alpha}}$ (that is almost -2 for large values of α) as t goes to infinity, while the variance decays exponentially, eventually reaching the constant value 0.08. Having this information, we can state that $Y(t)$ becomes stationary distributed around the attractor point $-2 + \frac{0.3816}{\sqrt{\alpha}}$ with variance 0.08 as $t \to \infty$

2.b

Now we consider the stochastic differential equation

$$dX(t) = -\alpha X(t)dt + \sqrt{\alpha}dW(t)$$

with $\alpha > 0$
We notice that the equation is similar to the one we solved before, since there $X(t)$ only multiplies the deterministic term, so we can solve it through the same method, with the integrating factor:

$$e^{\alpha t}dX(t) + \alpha e^{\alpha t}X(t)dt = e^{\alpha t}\sqrt{\alpha}dW(t)$$
$$\implies d(e^{\alpha t}X(t)) = e^{\alpha t}\sqrt{\alpha}dW(t)$$
$$\implies e^{\alpha t}X(t) = X(0) + \sqrt{\alpha}\int_0^t e^{\alpha s}dW(s)$$
$$\implies X(t) = e^{-\alpha t}X(0) + \sqrt{\alpha}e^{-\alpha t}\int_0^t e^{\alpha s}dW(s)$$

2.b.i

We compute expected value and variance of $X(t)$:

$$\mathbb{E}[X(t)] = e^{-\alpha t}X(0) + \sqrt{\alpha}e^{-\alpha t}\mathbb{E}[\int_0^t e^{\alpha s}dW(s)] = e^{-\alpha t}X(0)$$

Here we just used the linearity property of the expected value operator and that $\mathbb{E}[dW(s)] = 0$, from the definition of Wiener process.
So, taking the limit as t tends to infinity, we get:

$$\mathbb{E}[X(t)] \to 0$$

Now we compute the variance using the property:

$$\text{Var}(X(t)) = \mathbb{E}[X^2(t)] - \mathbb{E}[X(t)]^2$$

So we compute $\mathbb{E}[X^2(t)]$:

$$\mathbb{E}[X^2(t)] = e^{-2\alpha t}\mathbb{E}[X^2(0)] + \alpha e^{-2\alpha t}\mathbb{E}[(\int_0^t e^{\alpha s}dW(s))^2] + 2e^{-2\alpha t}\sqrt{\alpha}\mathbb{E}[X(0)\int_0^t e^{\alpha s}dW(s)] =$$

using the properties $\mathbb{E}[dW^2(s)] = ds$ and $\mathbb{E}[dW(s)] = 0$, we obtain:

$$= e^{-2\alpha t}\mathbb{E}[X^2(0)] + \alpha e^{-2\alpha t}\frac{e^{2\alpha t} - 1}{2\alpha}$$

So the variance is:

$$\text{Var}(X(t)) = e^{-2\alpha t}\mathbb{E}[X^2(0)] + \frac{1 - e^{-2\alpha t}}{2} - e^{-2\alpha t}\mathbb{E}^2[X(0)]$$

And its limit as $t \to \infty$:

$$\text{Var}(X(t)) \to \frac{1}{2}$$

2.b.ii

We start computing the forward Euler approximation:

$$X_{n+1} = (1 - \alpha\Delta t_n)X_n + \sqrt{\alpha}\Delta W_n$$

where $\Delta t_n = t_{n+1} - t_n$ and $\Delta W_n = W_{n+1} - W_n$
We can easily compute the expected value using the following recursion equation (that we got remembering that ΔW is normally distributed with zero mean and variance equal to Δt):

$$\mathbb{E}[X_{n+1}] = (1 - \alpha\Delta t)\mathbb{E}[X_n]$$

$$\mathbb{E}[X(0)] = X_0$$

$$\implies \mathbb{E}[X_n] = (1 - \alpha\Delta t)^n X_0$$

Thus, if we consider a uniform time discretisation with N points and $X_N = X(T)$, we have:

$$X(T) = (1 - \alpha\frac{T}{N})^N X_0$$

and its limit as t goes to ∞ ($\Leftrightarrow$ N goes to ∞) is:

$$X(T) = (1 - \alpha\frac{T}{N})^N X_0 \to e^{-\alpha T}X_0$$

Now we are interested in the variance. We can compute it similarly to what we have done before:

$$\mathrm{Var}(X_n) = \mathbb{E}[X_n^2] - \mathbb{E}^2[X_n]$$

After some straightforward computations (remembering that $\mathbb{E}[\Delta W_n] = 0$ and $\mathbb{E}[\Delta W_n^2] = \Delta t_n$)we got:

$$\mathbb{E}[X_{n+1}^2] = (1 - \alpha\Delta t_n)^2 \mathbb{E}[X_n^2] + \alpha\Delta t_n$$

$$\mathbb{E}[X_{n+1}]^2 = (1 - \alpha\Delta t_n)^2 \mathbb{E}[X_n]^2$$

Then we can write

$$\mathrm{Var}(X_{n+1}) = (1 - \alpha\Delta t_n)^2 \mathrm{Var}(X_n) + \alpha\Delta t_n$$

Iterating for large n (with a uniform time discretisation), we get:

$$\mathrm{Var}(X_\infty) = \frac{\alpha\Delta t}{1 - (1 - \alpha\Delta t)^2} = \frac{1}{2 - \alpha\Delta t}$$

2.b.iii

Now we do the same procedure but for the backward Euler scheme:

$$X_{n+1} = \frac{X_n}{1 + \alpha\Delta t_n} + \frac{\sqrt{\alpha}\Delta W_n}{1 + \alpha\Delta t_n}$$

Taking the expected value:

$$\mathbb{E}[X_{n+1}] = \frac{\mathbb{E}[X_n]}{1 + \alpha\Delta t_n}$$

Iterating (assuming $\mathbb{E}[X(0)] = X_0$ and a uniform time discretisation):

$$\mathbb{E}[X_n] = \frac{X_0}{(1 + \alpha\Delta t)^n}$$

Thus, the limit as $n \to \infty$ is:

$$\mathbb{E}[X_n] \to 0$$

Now we compute the variance using the same property as before:

$$\mathrm{Var}(X_n) = \mathbb{E}[X_n^2] - \mathbb{E}^2[X_n]$$

$$\mathbb{E}[X_{n+1}^2] = \frac{\mathbb{E}[X_n^2] + \alpha\Delta t}{(1 + \alpha\Delta t)^2}$$

Substituting the right terms, we get the following relation

$$\mathrm{Var}(X_{n+1}) = \frac{\mathrm{Var}(X_n)}{(1 + \alpha\Delta t)^2} + \frac{\alpha\Delta t}{1 + \alpha\Delta t)^2}$$

And iterating, we obtain:

$$\mathrm{Var}(X_\infty) = \frac{\alpha\Delta t}{(1 + \alpha\Delta t)^2 - 1} = \frac{1}{2 + \alpha\Delta t}$$

2.b.iv

Regarding (2.b.ii) and (2.b.iii) we can say that both the forward and backward Euler schemes converge to the right values of expected value and variance as $N \to \infty$ (notice that whenever $N \to \infty$ then $\Delta t \to 0$). This should be true since we define the solution of the Ito stochastic differential equation as the limit of forward Euler approximation, and the backward scheme is only affecting the deterministic term.
Moreover, we notice that the variance computed with backward Euler is always smaller (α ¿ 0 from hypothesis) then the one computed by forward Euler, they converge to the same value $\frac{1}{2}$.

2.c

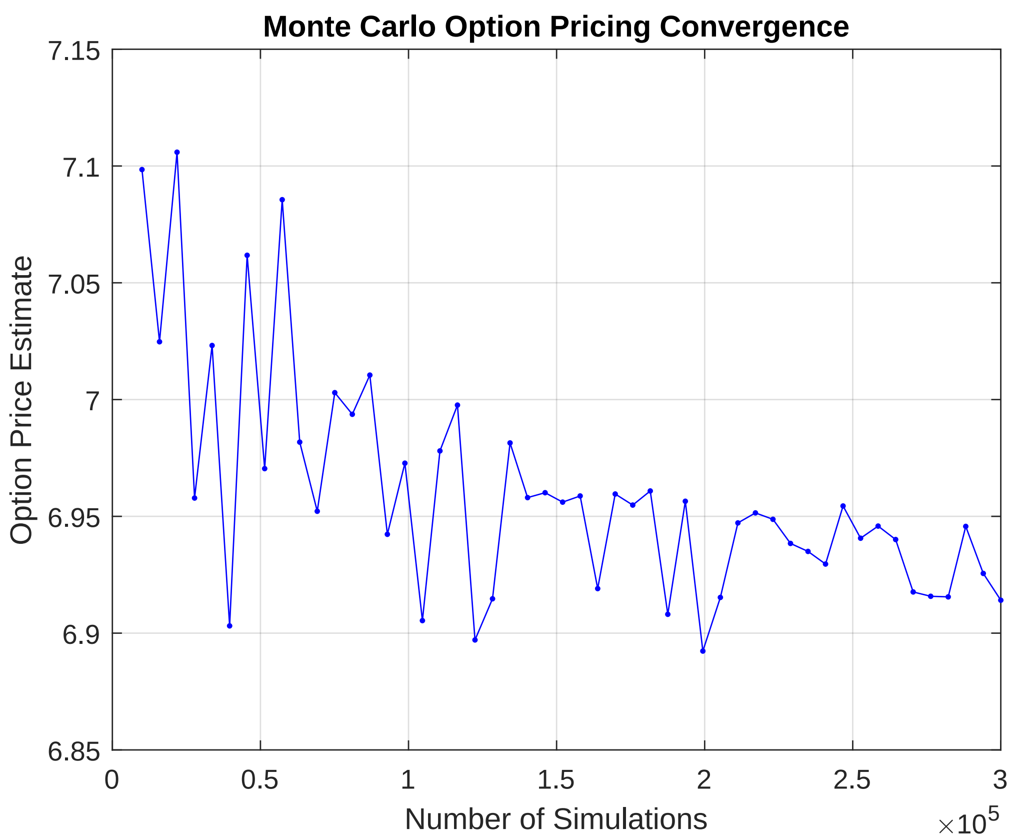
I computed the requested option value using the given forward Euler approximation in Matlab.
See code in file 2.c

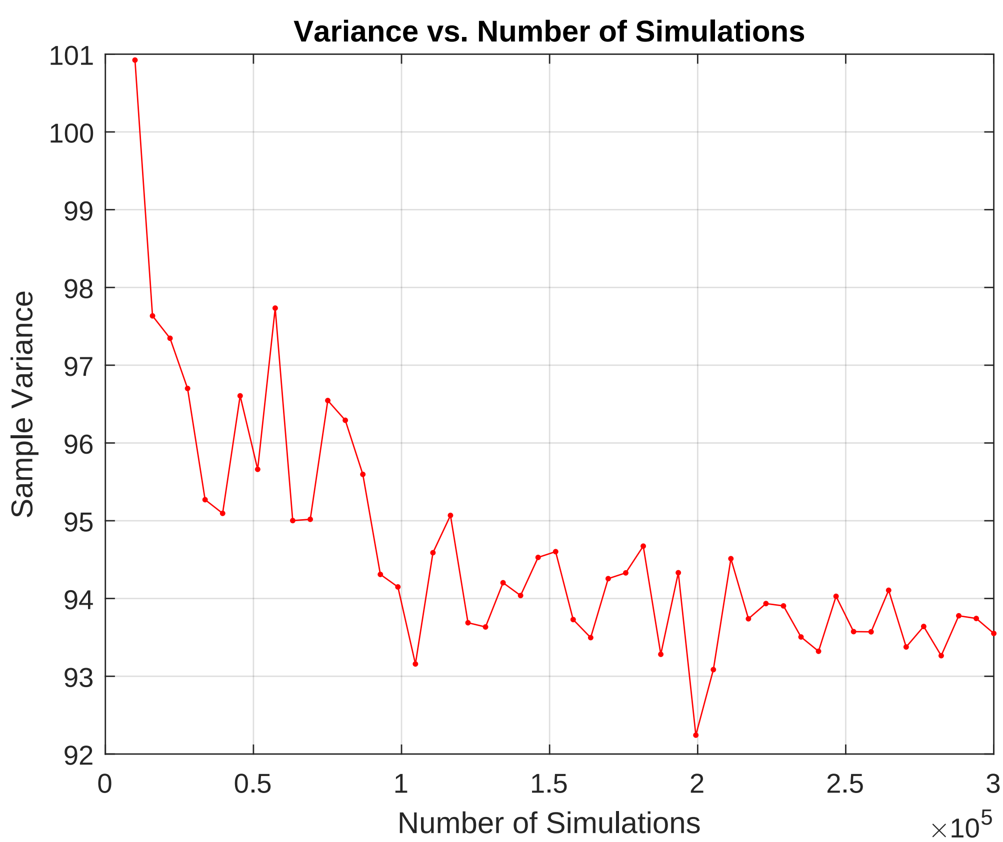
2.d

The Monte Carlo estimate has two kind of errors: the statistical error and the time discretisation error.
We defined the number of samples (simulations) as M and the number of points in the time discretisation in the interval as N. In the code I considered N smaller than M (so that the code was not too heavy), in particular I used $N = \sqrt{M}$.
To understand how many samples I needed to reach an accuracy of the values requested (0.05 and 0.005 later) I approximated the value of the Monte Carlo simulation error with

$$\frac{\hat{\sigma}}{\sqrt{M}}$$

(from point 1.a and the Central Limit Theorem), where $\hat{\sigma}$ is the square root of the sample variance. Then I plotted the values of the sample variance with respect to the number of samples M and took a value of sample variance on the basis of the value to which the graph approximately stabilized around for large values of M.


Monte Carlo Option Pricing Convergence
Option Price Estimate
Number of Simulations
7.15
7.1
7.05
7
6.95
6.9
6.85
0
0.5
1
1.5
2
2.5
3
×10^5



Variance vs. Number of Simulations
101
100
99
98
97
96
95
94
93
92
Sample Variance
0
0.5
1
1.5
2
2.5
3
Number of Simulations
×10^5

Looking at the graph, a reasonable value for the sample variance is $\hat{\sigma} = \sqrt{94}$ which seems to be more stable after $M = 2 * 10^5$. Taking these values, we get an error of

$$\frac{\hat{\sigma}}{\sqrt{M}} = \frac{\sqrt{94}}{\sqrt{2 * 10^5}} \approx 0.02$$

which is around the first tolerance 0.05.
If we want to satisfy the second tolerance (0.005), then we can find the necessary number of samples M with the following formula:

$$M = (\frac{\sqrt{94}}{0.005})^2 = 3.76 * 10^6$$